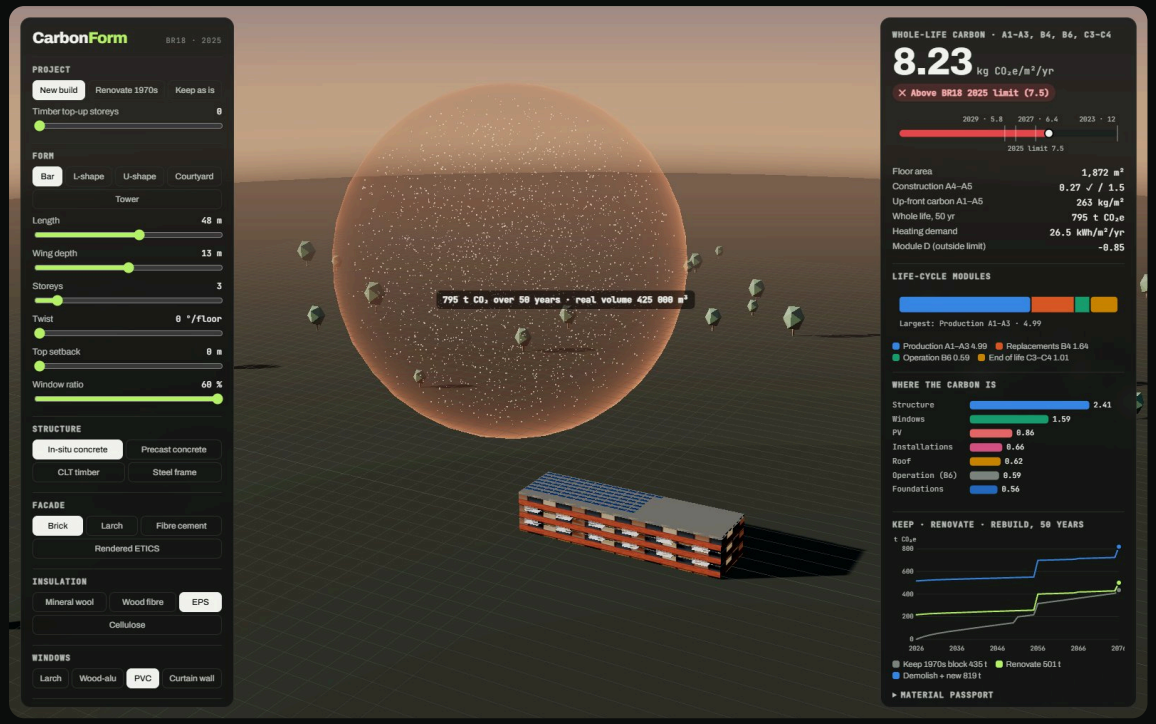
PARAMETRIC WHOLE-LIFE CARBON · BR18 2025 · DEMONSTRATION PROJECT

CarbonForm



Shape a building, swap its materials and renovate it, and its whole-life carbon updates live against the Danish BR18 limit. The 50-year emissions appear next to it as a cloud of CO_2 gas at its true volume.

6.47
CONCRETE + BRICK

3.86
CLT + LARCH

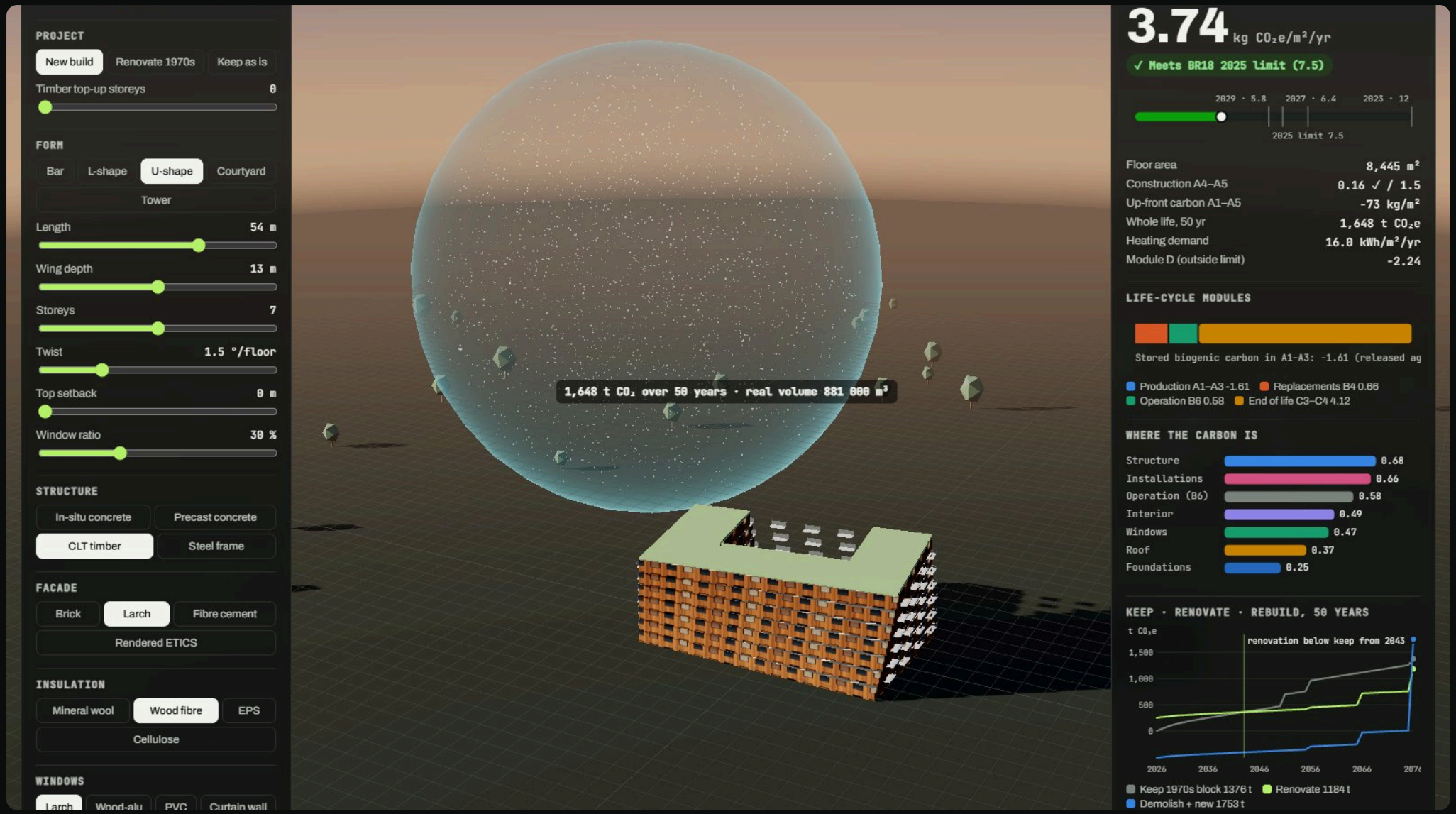
2.80
RENOVATE 1970S

Data: **BR18 annex 2 table 7 (2025)** (33 datasets) and the BR18 installation standard values, the official 2025–2075 Danish emission factors for electricity and district heating, and ÖKOBAUDAT for two gaps. Exports a project to **LCAbyg**.

This low-rise glazed concrete scheme scores **8.23**, **above the 7.5 limit**, so the cloud turns orange.

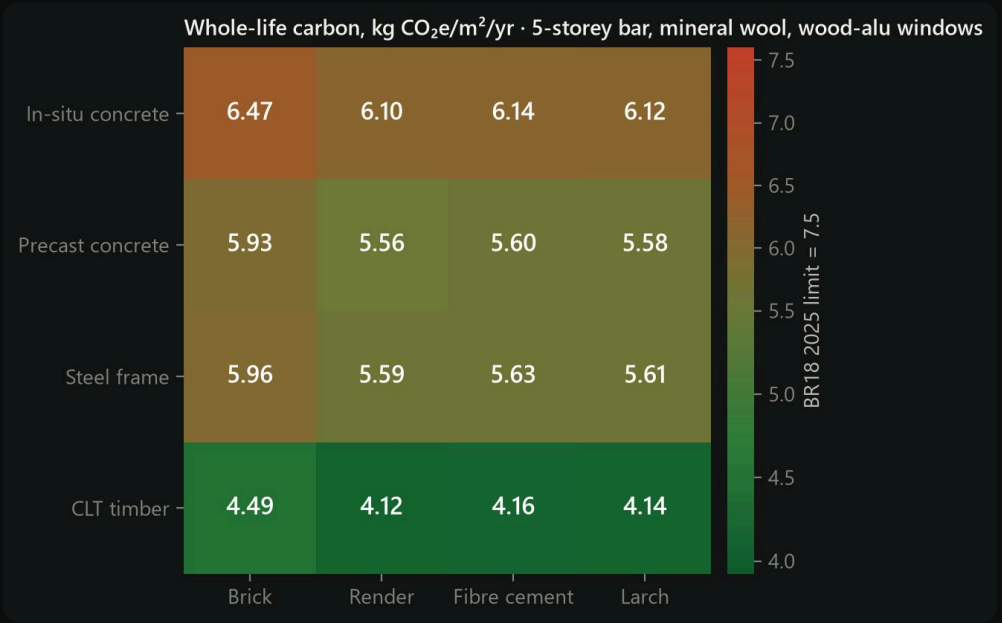
THE TOOL

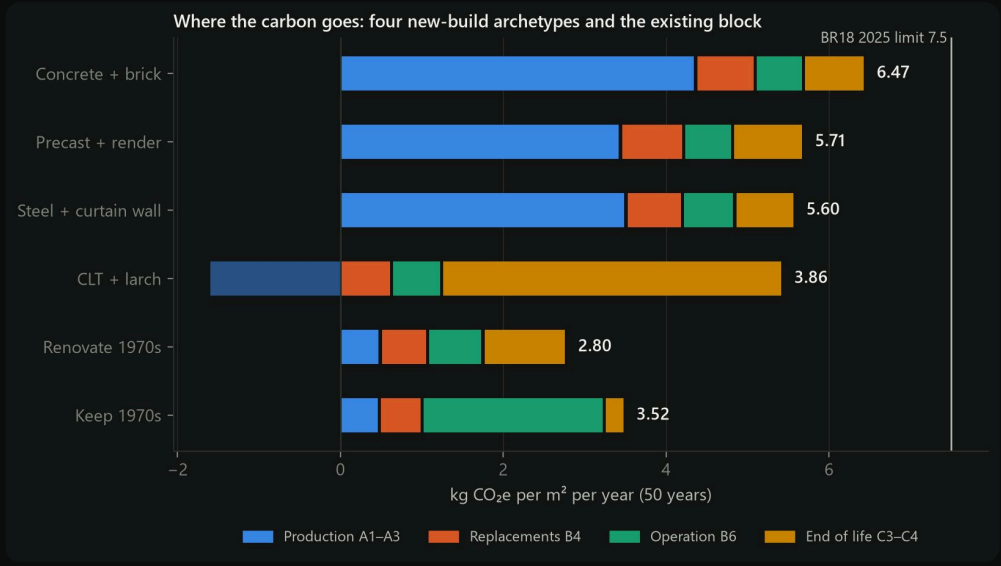
Every slider is a design decision; every decision has a number



RESULTS

Structure decides; the facade fine-tunes

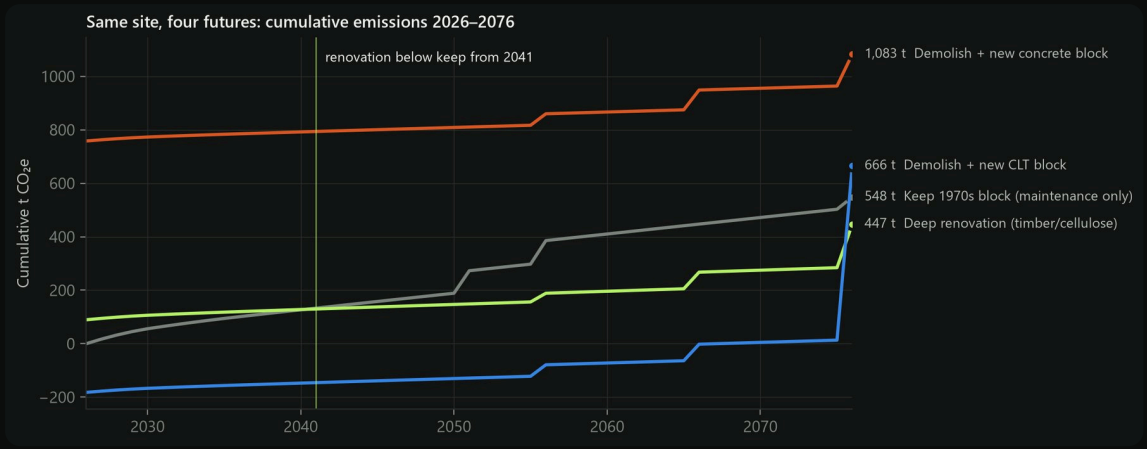




Per m² per year over 50 years, A1–A3 + B4 + B6 + C3–C4. Timber shows its stored (biogenic) carbon as negative production and releases it at end of life (EN 15804+A2). Every new-build archetype passes 7.5; in-situ concrete with brick is closest to the limit.

RENOVATION

The greenest building is often the one already standing



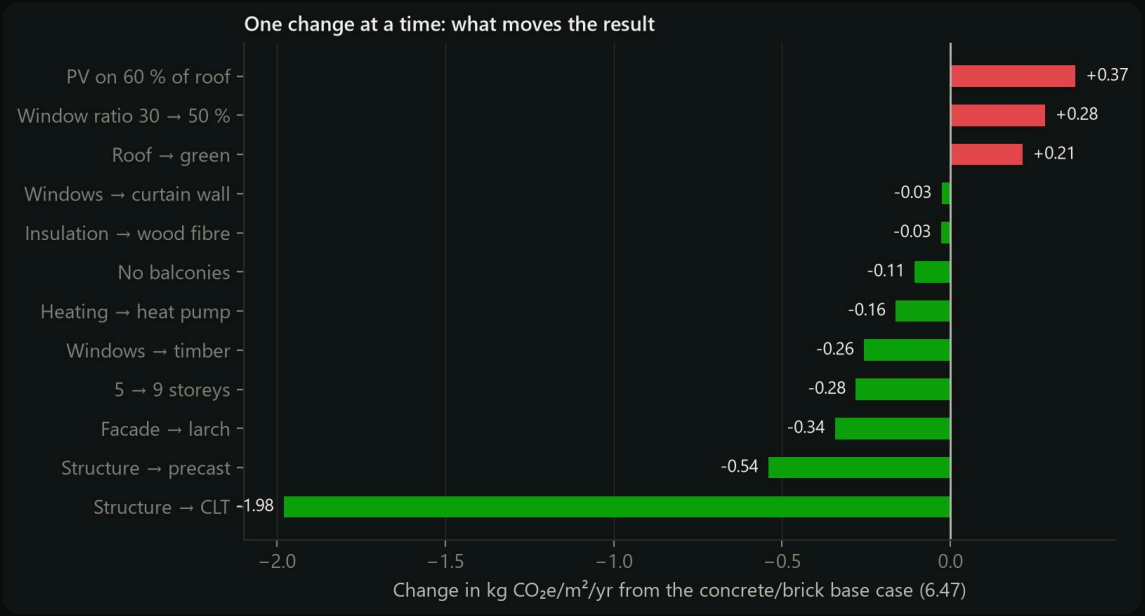
Same site, 50 years. Keeping the 1970s block still costs heating (≈130 kWh/m² a year) plus ordinary maintenance. A deep renovation cuts heating to ≈20 kWh/m² for a fraction of a new building's materials. Demolish-and-rebuild starts with the old block's end-of-life and a new building's production emissions.

89 t
RENOVATION UP-FRONT

131 → 23
KWH/M² HEATING

SENSITIVITY

What actually moves the result



- **Structure is the big lever.** CLT instead of in-situ concrete is −1.98; precast is −0.54.
- **Facade and windows are second order** (−0.3 each).
- **Rooftop PV adds carbon under these rules** (+0.37). With Denmark's 2025–2075 grid factors (≈0.025 kg/kWh) and PV credited only against the building's own use, the panels' production outweighs the saved electricity.
- **Taller is better per m²:** 5 → 9 storeys is −0.28, because foundations and roof are shared by more floor area.

DATA, METHOD AND VERIFICATION

Built on the regulation's own numbers

Material	Table 7 ID	Unit	A1–A3
Ready-mix concrete C30/37	#B1477	m^3	282
Ready-mix concrete C25/30 (foundations, toppings)	#B1433	m^3	241
Reinforcing steel	#D1590	kg	1.14
Structural steel, rolled sections	#D1591	kg	1.64
Hollow-core slab element, 22 cm	#B1346	m^2	49.8
Precast concrete wall element, 20 cm	#B1389	m^2	66.9
Precast concrete wall element, 15 cm	#B1385	m^2	53.1
Cross-laminated timber (CLT)	#B1318	m^3	-664
Glulam, spruce/pine	#B1339	m^3	-610
Structural timber, sawn and dried	#B1489	m^3	-680
Larch (cladding)	#G0221	m^3	-903
Facing brick (yellow/brown)	#D1508	kg	0.265
Fibre-cement board	#D1514	m^2	12.9
ETICS adhesive + silicate render	#G1251	m^2	5.55
Mineral wool, hard, class 34	#D1536	m2R	1.24
Mineral wool, hard, class 37	#D1537	m2R	1.5
Wood-fibre insulation board	#D1627	m2R	-2.14
EPS insulation, grey, 60 kPa	#D1560	m2R	1.52
Cellulose (paper wool) insulation	#D1623	kg	-1.19
Timber window, triple glazed	#D1635	m^2	51.7
Timber-aluminium window, triple glazed	#D1634	m^2	90.6
PVC window, triple glazed	#D1632	m^2	101
Curtain wall, steel, triple glazed	#G0111	m^2	95.9
Roofing felt, top + base layer	#D1501	m^2	5.69

Rule / check	Value
BR18 2025 limit, multi-storey housing	7.5
A4–A5 construction limit	1.5
Reference period	50 yr
B6 factors electricity / district heating 2025 → 2050	0.080→0.025 / 0.042→0.013
Installations (BR18 standard values, housing)	0.50 A1–A3
JS vs Python engine, 300 random designs	agree to 1e-9
LCAbyg export, recomputed with LCAbyg rules	exact, 40 designs
Automated tests	19 / 19
Heating demand new / 1970s	17 / 131 kWh/m²

Limits. Parametric quantities (no basement, lift cores or detailed finishes); generic data; screening energy model (not Be18); a design-stage tool, not a BR18 submission. The LCAbyg export is the bridge to an official calculation.

Sources: bygningsreglementet.dk (bilag 2, tabel 7, 2025) · SBST emissionsfaktorer 2025–2075 · oekobaudat.de · LCAbyg JSON guide